MOHAMMAD OMAR KHURSHEED

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I am an AI Safety researcher working on AI alignment with 5 years of experience shipping speech models at Amazon and a research background in machine learning and computational social science.

ALIGNMENT & SOCIETAL IMPACTS EXPERIENCE

Anthropic (via MATS) — *Berkeley, CA*
AI Safety Fellow (Mentor: Fabien Roger) — *January 2026 – Present*

- Studying if **helpful-only training** produces models with different **censorship, refusal, and misalignment properties** than HHH-style baselines, using 8B to 32B models.
- Developed training pipelines using SFT, Synthetic Document Fine-tuning, and GRPO to build helpful-only model organisms and compare them against HHH counterparts.
- Developed evals to **measure misalignment and capabilities** across axes including reasoning ability, quality of CoT, and non-refusal. Found that character training is a required component to develop helpful-only models that are not misaligned.

Social Insight Lab (Prof. Ashiqur Rahman KhudaBukhsh), RIT — *Remote*
Research Collaborator — *July 2025 – Present*

- Built an LLM-as-judge pipeline to classify stance and sexism across 6,531 **suffrage-related speeches from 200+ years of UK parliamentary debate**, validated against expert annotation ($\kappa = 0.489$, matching human-human agreement); paper in submission at ACL Rolling Review.
- **Showed that opposing stances deploy different rhetorical structures of sexism**: hostile/proscriptive among opponents versus benevolent/paternalistic among supporters, providing naturalistic evidence for social-psychological theories of misogyny across two centuries of institutional discourse.
- Released a 6.7M-speech gender-matched Hansard dataset (89% speaker-gender coverage across the **House of Commons**) and demonstrated that careful LLM judge design grounded in established social-psychological taxonomies can yield reliable insights on subjective and temporally varied data.

NGRAM Lab (Prof. Mohit Iyyer), UMass Amherst — *Amherst, MA*
Graduate Student Researcher — *January – December 2020*

- Curated a large-scale dataset linking narrative tropes with IMDb and Goodreads metadata to study gender bias in narrative fiction; co-developed a **"genderedness" score** and studied its correlation with ratings and creator gender (published at EMNLP NLP+CSS Workshop).

SELECTED PUBLICATIONS

- (Mis)generalization of Helpful-only Fine-tuning. **Khursheed, MO***, Sosis B*, Roger F (*in submission at NeurIPS*), 2026
- Two Centuries of Women's Voices: Computational Analysis of Gender Participation and Suffrage Debates in British Parliament. **Khursheed MO***, Sawkar M*, Khudabukhsh AR (*in submission at ARR*), 2026.
- Small Footprint Slimmable Networks for Keyword Spotting. Akhtar Z, **Khursheed MO**, Du D, Liu Y. *ICASSP*, 2023.
- Latency Control for Keyword Spotting. Jose C, Wang J, Strimel GP, **Khursheed MO**, Mishchenko Y, Kulis B. *Interspeech*, 2022.
- Tiny-CRNN: Streaming Wakeword Detection in a Low Footprint Setting. **Khursheed MO***, Jose C, Kumar R, Fu G, Kulis B, Cheekatmalla SK. *ASRU*, 2021.
- Analyzing Gender Bias within Narrative Tropes. Gala D*, **Khursheed MO***, Lerner H, O'Connor B, Iyyer M. *NLP+CSS Workshop @ EMNLP*, 2020.

INDUSTRY EXPERIENCE

Amazon – Alexa Edge AI — *Sunnyvale, CA*
Applied Scientist II — *December 2022 – December 2025*

- Developed systematic evaluation frameworks for **measuring demographic fairness gaps** in wakeword detection across gender, accent, and regional cohorts; designed targeted training strategies that achieved 10% relative FRR reduction for underperforming groups on Echo devices.
- Led a project to **predict device-directedness using late fusion of audio metadata and neural network posteriors**, reducing false wakes by 50% during natural conversation and requiring careful analysis of failure modes and robustness under distribution shift.
- Designed a dynamically slimmable transformer architecture for wakeword detection on low-power devices, enabling systematic study of compute/accuracy tradeoffs across deployment constraints.

Applied Scientist I *January 2021 – December 2022*

- Led a team to develop low-power wakeword models for mobile and earbuds, improving previous models by up to **60% relative FRR** across multiple locales using semi-supervised learning on large unlabeled datasets.
- Shipped production models to 10M+ users with calibrated thresholds and monitoring; received the Amazon Deliver Results Award (Q1–Q2 2021).

Applied Scientist Intern *May 2020 – September 2020*

- Developed small-footprint CRNNs for keyword spotting, improving accuracy over **CNNs by ∼25% with ∼10% fewer parameters**; published at ASRU 2021.

EDUCATION

University of Massachusetts Amherst *2019 – 2021*
M.S. in Computer Science GPA: 3.97/4.0

Aligarh Muslim University, Aligarh *2015 – 2019*
B.Tech. in Computer Engineering GPA: 9.52/10.0
Selected achievements: Sir Syed Global Scholar Award; ICPC Regionals; AMU-OSS Co-founder; Hack36 2018 Top 12.

SERVICE

- Reviewer: NeurIPS 2024; ICML 2025; AAAI 2026; AMLC 2023–2025; SIGIR 2022; JMIR